

JOHN M. REKOSKE, PH.D.

john.rekoske@gmail.com | 920-228-0173 | Pasadena, CA

PROFESSIONAL SUMMARY

Computational geophysicist specializing in physics-aware machine learning and reduced-order modeling (ROM) for seismological applications. Demonstrated expertise designing surrogate and ROM frameworks that accelerate high-fidelity physics simulations (elastic wave propagation, seismic hazard analysis) by orders of magnitude, enabling real-time inference on large-scale problems. Strong scientific record with 15 peer-reviewed publications and substantial experience with HPC environments, deep learning, and scientific software development in Python, C++, and Fortran. Proven ability to bridge AI/ML research and domain science communities, with experience collaborating across geophysics, engineering, and computational science.

EDUCATION

University of California, San Diego, La Jolla, CA — Ph.D. Geophysics, 2026

Colorado School of Mines, Golden, CO - B.S. Geophysical Engineering, 2020

WORK EXPERIENCE

Caltech Seismological Laboratory Director's Postdoctoral Research Associate in Geophysics, California Institute of Technology; Pasadena, CA — 2026-Present

Leads research of next-generation earthquake early warning algorithms and methodology using surrogate models, large heterogeneous datasets, and physics-aware machine learning to support real-time, operational warning and monitoring for the safety of critical infrastructure.

National Science Foundation Graduate Research Fellow in Geophysics, Scripps Institution of Oceanography; La Jolla, CA — 2021-2026

Conducted geophysical research on earthquake ground motions for earthquake engineering using machine learning and novel reduced-order models to accelerate numerical elastic wave propagation simulations for earthquake early warning, seismic hazard analysis, and to better understand earthquake source and rupture processes. Published 3 first-author research articles, co-authored 4 research articles, and published 3 seismic datasets.

Physical Scientist of Earthquake Hazards, United States Geological Survey; Golden, CO — 2018-2021

Core developer of official USGS software where I developed a research prototype into production-grade operational software deployed for national seismic monitoring and seismic data workflows including ML-based inverse methods for ground motion parameter estimation. Compiled datasets of earthquake ground motion data and analyzed ground motion models to improve understanding of shaking and structural dynamics.

SKILLS

- Scientific Computing & HPC: Python (NumPy, SciPy, Pandas, Matplotlib, ObsPy), C/C++, Fortran, R, SQL; MPI/OpenMP parallel workflows; TACC Frontera, LRZ SuperMUC-NG; SDSC Expanse; HDF5, Conda, Git
- AI/ML Frameworks & Methods: PyTorch, TensorFlow/Keras, Scikit-learn; physics-informed neural networks (PINNs); reduced-order modeling (ROM/POD); surrogate modeling; convolutional neural networks (CNNs); inverse problem methods; uncertainty quantification; V&V of ML models for physical consistency; feature engineering for time-series/waveform data
- Physics Simulation Platforms: elastic wave propagation; finite element modeling; discontinuous Galerkin method; ground motion simulation; seismic hazard analysis, SeisSol, SPECFEM

AWARDS

- Caltech Seismological Laboratory Director's Fellowship
- National Science Foundation Graduate Research Fellowship
- Scripps Institution of Oceanography Institutional Fellowship
- Seismological Society of America Travel Grant Award
- Colorado School of Mines Cecil H. Green Gold Medal Award

PUBLICATIONS (H-INDEX=12)

1. **Rekoske, J. M.**, Callaghan, S., Milner, K., May, D., and Gabriel, A.-A (2026). CyberShake physics-based seismic hazard maps using reduced-order models: results from Southern San Andreas earthquake scenarios. *Bulletin of the Seismological Society of America*.
2. **Rekoske, J. M.**, May, D. A., and Gabriel, A.-A. (2025). Reduced-order modeling for complex 3D seismic wave propagation. *Geophysical Journal International* 241, no. 1, 528-548.
3. Thompson, E., Hearne, M., Aagaard, B. T., **Rekoske, J. M.**, Worden, C. B., Moschetti, M. P., ... and Kottke, A. R. (2024). Automated, Near Real-Time Ground-Motion Processing at the US Geological Survey. *Seismological Research Letters*.
4. **Rekoske, J. M.**, Gabriel, A.-A., & May, D. A. (2023). Instantaneous Physics-Based Ground Motion Maps Using Reduced-Order Modeling. *Journal of Geophysical Research: Solid Earth* 128, no. 8, e2023JB026975.
5. Gabriel, A.-A., T. Ulrich, M. Marchandon, J. Biemiller, and **Rekoske, J. M.** (2023). 3D Dynamic Rupture Modeling of the 6 February 2023, Kahramanmaraş, Turkey M w 7.8 and 7.7 Earthquake Doublet Using Early Observations. *The Seismic Record* 3 no. 4, 342-356.
6. Jia, Z., Z. Jin, M. Marchandon, T. Ulrich, A.-A. Gabriel, W. Fan, P. Shearer, X. Zou, **Rekoske, J. M.**, F. Bulut, et al. (2023). The complex dynamics of the 2023 Kahramanmaraş, Turkey, Mw 7.8-7.7 earthquake doublet, *Science* 381, no. 6661, 985-990.
7. Nayak, A., V. Rodríguez Tribaldos, J. Ajo-Franklin, B. Miranda, C.-C. Chien, R. Mellors, M. Robertson, M. Brandin, **Rekoske, J. M.**, and T. Wood (2023). Nodal and Broadband Seismometer Complement to the Imperial Valley Dark Fiber DAS Array, *Seismological Research Letters* 94, no. 6, 2852-2867.
8. **Rekoske, J. M.**, M. P. Moschetti, and E. M. Thompson (2022). Basin and site effects in the US Pacific Northwest estimated from small-magnitude earthquakes, *Bulletin of the Seismological Society of America* 112, no. 1, 438-456.
9. Kleckner, J. K., K. B. Withers, E. M. Thompson, **Rekoske, J. M.**, E. Wolin, and M. P. Moschetti (2022). Automated detection of clipping in broadband earthquake records, *Seismological Research Letters* 93, no. 2A, 880-896.
10. Moschetti, M. P., D. Churchwell, E. M. Thompson, **Rekoske, J. M.**, E. Wolin, and O. S. Boyd (2021). Seismic wave propagation and basin amplification in the Wasatch Front, Utah, *Seismological Society of America* 92, no. 6, 3626-3641.
11. **Rekoske, J. M.**, E. M. Thompson, M. P. Moschetti, M. G. Hearne, B. T. Aagaard, and G. A. Parker (2020). The 2019 Ridgecrest, California, earthquake sequence ground motions: Processed records and derived intensity metrics, *Seismological Research Letters* 91, no. 4, 2010-2023.
12. Moschetti, M. P., E. M. Thompson, **Rekoske, J. M.**, M. G. Hearne, P. M. Powers, D. E. McNamara, and C. Tape (2020). Ground-Motion amplification in Cook Inlet Region, Alaska, from intermediate-depth earthquakes, including the 2018 Mw 7.1 Anchorage earthquake, *Seismological Research Letters* 91, no. 1, 142-152.
13. Parker, G. A., A. S. Baltay, **Rekoske, J. M.**, and E. M. Thompson (2020). Repeatable source, path, and site effects from the 2019 M 7.1 Ridgecrest earthquake sequence, *Bulletin of the Seismological Society of America* 110, no. 4, 1530-1548.
14. McNamara, D. E., Wolin, E., Powers, P. M., Shumway, A. M., Moschetti, M. P., **Rekoske, J. M.**, ... & Petersen, M. D. (2020). Evaluation of Ground-Motion Models for US Geological Survey Seismic Hazard Models: 2018 Anchorage, Alaska, Mw 7.1 Subduction Zone Earthquake Sequence, *Seismological Research Letters* 91, no. 1, 183-194.
15. McNamara, D. E., Wolin, E., Powers, P. M., Shumway, A. M., Moschetti, M. P., **Rekoske, J. M.**, ... & Petersen, M. D. (2020). Evaluation of ground-motion models for US Geological Survey seismic hazard forecasts: Hawaii tectonic earthquakes and volcanic eruptions. *Bulletin of the Seismological Society of America* 110, no. 2, 666-688.

PUBLISHED GEOPHYSICAL DATASETS

1. **Rekoske, J. M.**, May, D. A., & Gabriel, A.-A. (2024). Dataset for “Reduced-order modeling for complex 3D seismic wave propagation”. Zenodo.
2. **Rekoske, J. M.**, Gabriel, A.-A., & May, D. A. (2023). jrekoske/Instantaneous-PhysicsBased-GroundMotionMaps: Publication release (v1.0.0). Zenodo.
3. **Rekoske, J. M.**, Moschetti, M. P., & Thompson, E. M. (2019). Database of ground motions from tectonic and volcanic events, Hawaii, 2018. Center for Engineering Strong Motion Data (CESMD).
4. **Rekoske, J. M.**, Thompson, E. M., Moschetti, M. P., Hearne, M. G., Aagaard, B. T., & Parker, G. A. (2019). Ground motions from the 2019 Ridgecrest, California, earthquake sequence. Center for Engineering Strong Motion Data (CESMD).
5. **Rekoske, J. M.**, Moschetti, Morgan P., Thompson, Eric M and Frankel, Arthur D. (2021). Database of horizontal-component Fourier amplitude spectra of acceleration ground motions from Pacific Northwest earthquakes, 1999-2020. Center for Engineering Strong Motion Data (CESMD).
6. Churchwell, David, Moschetti, Morgan P., Thompson, Eric M., **Rekoske, J. M.**, Wolin, Emily, and Boyd, Oliver (2021). Database of horizontal-component Fourier amplitude spectra of acceleration ground motions from Utah earthquakes, 2010-2020. Center for Engineering Strong Motion Data (CESMD).
7. **Rekoske, J. M.**, Morgan P. Moschetti, Eric M. Thompson, Mike G. Hearne, Carl Tape (2019). USGS database of ground motions from in-slab earthquakes near Anchorage, Alaska, 2008-2019. Center for Engineering Strong Motion Data (CESMD).

INVITED TALKS AND SEMINARS

1. **Rekoske, J. M.**, L. Kuratle, D. Kilb, & A.-A. Gabriel. Improving Scientific Communication using Earthquake Simulations: Insights from A Public Survey. American Geophysical Union Annual Meeting. December 2025. New Orleans, LA.
2. **Rekoske, J. M.**, A.-A. Gabriel, & D. A. May. Rapid physics-informed wavefields using high-performance computing and reduced-order models. United States Geological Survey, Earthquake Science Center Seminar. September 2024. Moffett Field, CA.
3. **Rekoske, J. M.**, A.-A. Gabriel, & D. A. May. Rapid physics-informed wavefields using high-performance computing and reduced-order models. Megathrust Modeling Framework, All-hands Science Discussion. October 2024.